

Michiel A. P. Pertijs

associate professor

基本信息

姓名: Michiel A. P. Pertijs

所属院校: 代尔夫特理工大学

地区: Netherlands

年龄范围: 50岁以上

海外经历: 有

企业经历: 有

学科领域

生物医学工程

集成电路设计与集成系统

研究领域

医学影像处理与成像技术

混合信号集成技术

生物医学传感与检测技术

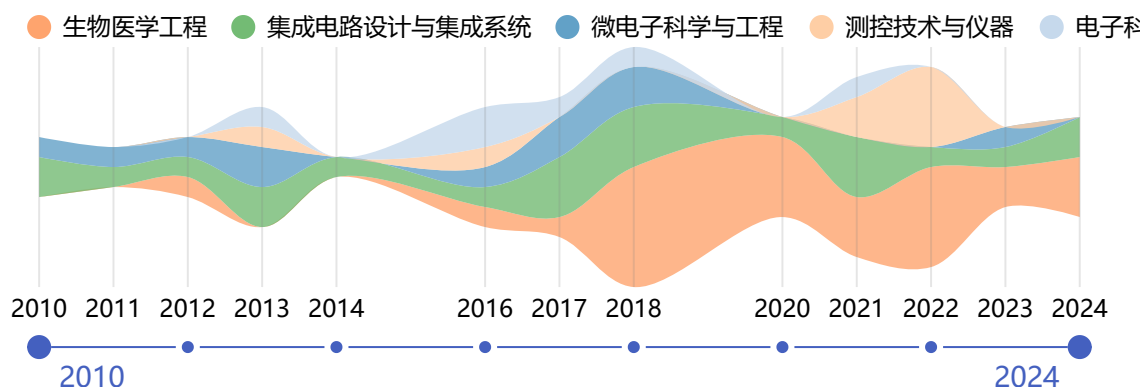
低功耗电路与能量管理

ASIC与可编程器件

个人简介

Michiel A. P. Pertijs is an Associate Professor in the Electronic Instrumentation (EI) of the Department of Microelectronics. He holds M.Sc. and Ph.D. degrees in electrical engineering from Delft University of Technology. His expertise includes sensor interface circuits, smart sensors, etc. He heads a research group on integrated circuits for medical ultrasound and energy - efficient smart sensors. Dr. Pertijs has authored or co - authored 2 books, 3 book chapters, 13 patents, and over 100 papers. He served as an Associate Editor of IEEE JSSC and is on the technical program committee of ESSCIRC. He has won several awards and teaches multiple courses at Delft University of Technology.

近年发表的论文



An Amplitude-Programmable Energy-Recycling High-Voltage Resonant Pulser for Battery-Powered Ultrasound Devices

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bellouki, Imad; Rozsa, Nuriel N. M.; Chang, Zu-Yao; Chen, Zhao; Tan, Mingliang; Pertijs, Michiel A. P.

被引用次数: 0

2024

A 125 μ m-Pitch-Matched Transceiver ASIC With Micro-Beamforming ADC and Multi-Level Signaling for 3-D Transfontanelle Ultrasonography

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Guo, Peng; Fool, Fabian; Chang, Zu-Yao; Noothout, Emile; Vos, Hendrik J.; Bosch, Johan G.; de Jong, Nico; Verweij, Martin D.; Pertijs, Michiel A. P.

被引用次数: 1

2024

A Pitch-Matched High-Frame-Rate Ultrasound Imaging ASIC for Catheter-Based 3-D Probes

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Hopf, Yannick M.; dos Santos, Djalma Simoes; Ossenkoppele, Boudewine W.; Soozande, Mehdi; Noothout, Emile; Chang, Zu-Yao; Chen, Chao; Vos, Hendrik J.; Bosch, Johan G.; Verweij, Martin D.; de Jong, Nico; Pertijs, Michiel A. P.

被引用次数: 6

2024

An Ultrasound Matrix Transducer for High-Frame-Rate 3-D Intra-cardiac Echocardiography

ULTRASOUND IN MEDICINE AND BIOLOGY

dos Santos, Djalma Simoes; Ossenkoppele, Boudewine; Hopf, Yannick M.; Soozande, Mehdi; Noothout, Emile; Vos, Hendrik J.; Bosch, Johan G.; Pertijs, Michiel A. P.; Verweij, Martin D.; de Jong, Nico

被引用次数: 3

2024

A 1.2-mW/Channel Pitch-Matched Transceiver ASIC Employing a Boxcar-Integration-Based RX Micro-Beamformer for High-Resolution 3-D Ultrasound Imaging

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Guo, Peng; Fool, Fabian; Chang, Zu-Yao; Noothout, Emile; Vos, Hendrik J.; Bosch, Johan G.; de Jong, Nico; Verweij, Martin D.; Pertijs, Michiel A. P.

被引用次数: 1

2023

A Pitch-Matched Low-Noise Analog Front-End With Accurate Continuous Time-Gain Compensation for High-Density Ultrasound Transducer Arrays

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Guo, Peng; Chang, Zu-Yao; Noothout, Emile; Vos, Hendrik J.; Bosch, Johan G.; de Jong, Nico; Verweij, Martin D.; Pertijs, Michiel A. P.

被引用次数: 10

2023

Imaging Scheme for 3-D High-Frame-Rate Intracardiac Echography: A Simulation Study

IEEE TRANSACTIONS ON ULTRASONICS FERROELECTRICS AND FREQUENCY CONTROL

Soozande, Mehdi; Ossenkoppele, Boudewine W.; Hopf, Yannick; Pertijs, Michiel A. P.; Verweij, Martin D.; de Jong, Nico; Vos, Hendrik J.; Bosch, Johan G.

被引用次数: 4

2022

A Pitch-Matched Transceiver ASIC With Shared Hybrid Beamforming ADC for High-Frame-Rate 3-D Intracardiac Echocardiography

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Hopf, Yannick M.; Ossenkoppele, Boudewine W.; Soozande, Mehdi; Noothout, Emile; Chang, Zu-Yao; Chen, Chao; Vos, Hendrik J.; Bosch, Johan G.; Verweij, Martin D.; de Jong, Nico; Pertijs, Michiel A. P.

被引用次数: 10

2022

Design and Proof-of-Concept of a Matrix Transducer Array for Clamp-On Ultrasonic Flow Measurements

IEEE TRANSACTIONS ON ULTRASONICS FERROELECTRICS AND FREQUENCY CONTROL

Massaad, Jack; Van Neer, Paul L. M. J.; Van Willigen, Douwe M.; Noothout, Emile C.; de Jong, Nicolaas; Pertijs, Michiel A. P.; Verweij, Martin D.

被引用次数: 2

2022

Measurement of Pipe and Fluid Properties With a Matrix Array-Based Ultrasonic Clamp-On Flow Meter

IEEE TRANSACTIONS ON ULTRASONICS FERROELECTRICS AND FREQUENCY CONTROL

Massaad, Jack; van Neer, Paul L. M. J.; van Willigen, Douwe M.; Sabbadini, Alberico; de Jong, Nicolaas; Pertijs, Michiel A. P.; Verweij, Martin D.

被引用次数: 9

2022

Automated Characterization of Matrix Transducer Arrays using the Verasonics Imaging System

2022 IEEE INTERNATIONAL ULTRASONICS SYMPOSIUM (IEEE IUS)

dos Santos, Djalma Simoes; Fool, Fabian; Kim, Taehoon; Noothout, Emile; Rozsa, Nuriel; Vos, Hendrik J.; Bosch, Johan G.; Pertijs, Michiel A. P.; Verweij, Martin D.; de Jong, Nico

被引用次数: 1

2022

Large Matrix Array Aperture for 3D Vascular Imaging Capture

2022 IEEE INTERNATIONAL ULTRASONICS SYMPOSIUM (IEEE IUS)

Colas, Quorentin; Bantignies, Claire; Perroteau, Marie; Porcher, Nicolas; Vassal, Steeven; Guerif, Benjamin; Kim, Taehoon; Bosch, Johan G.; de Jong, Nico; Verweij, Martin D.; Pertijs, Michiel A. P.; Ferin, Guillaume; Flesch, Martin

A Tiled Ultrasound Matrix Transducer for Volumetric Imaging of the Carotid Artery

SENSORS

dos Santos, Djalma Simoes; Fool, Fabian; Mozaffarzadeh, Moein; Shabanimotlagh, Maysam; Noothout, Emile; Kim, Taehoon; Rozsa, Nuriel; Vos, Hendrik J.; Bosch, Johan G.; Pertijs, Michiel A. P.; Verweij, Martin D.; de Jong, Nico

被引用次数: 2

2022

Algorithm to Correct Measurement Offsets Introduced by Inactive Elements of Transducer Arrays in Ultrasonic Flow Metering

SENSORS

Massaad, Jack; van Neer, Paul L. M. J.; van Willigen, Douwe M.; Pertijs, Michiel A. P.; de Jong, Nicolaas; Verweij, Martin D.

被引用次数: 0

2022

Measurement of Pipe and Liquid Parameters Using the Beam Steering Capabilities of Array-Based Clamp-On Ultrasonic Flow Meters

SENSORS

Massaad, Jack; van Neer, Paul L. M. J.; van Willigen, Douwe M.; Pertijs, Michiel A. P.; de Jong, Nicolaas; Verweij, Martin D.

被引用次数: 3

2022

An Algorithm to Minimize the Zero-Flow Error in Transit-Time Ultrasonic Flowmeters

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

van Willigen, Douwe M.; van Neer, Paul L. M. J.; Massaad, Jack; de Jong, Nico; Verweij, Martin D.; Pertijs, Michiel A. P.

被引用次数: 12

2021

Design of an Ultrasound Transceiver ASIC with a Switching-Artifact Reduction Technique for 3D Carotid Artery Imaging

SENSORS

Kim, Taehoon; Fool, Fabian; dos Santos, Djalma Simoes; Chang, Zu-Yao; Noothout, Emile; Vos, Hendrik J.; Bosch, Johan G.; Verweij, Martin D.; de Jong, Nico; Pertijs, Michiel A. P.

被引用次数: 8

2021

Experimental Investigation of the Effect of Subdicing on an Ultrasound Matrix Transducer

INTERNATIONAL ULTRASONICS SYMPOSIUM (IEEE IUS 2021)

dos Santos, Djalma Simoes; Fool, Fabian; Kim, Taehoon; Noothout, Emile; Vos, Hendrik J.; Bosch, Johan G.; Pertijs, Michiel A. P.; Verweij, Martin D.; de Jong, Nico

被引用次数: 3

2021

A Transceiver ASIC for a Single-Cable 64-Element Intra-Vascular Ultrasound Probe

IEEE JOURNAL OF SOLID-STATE CIRCUITS

van Willigen, Douwe M.; Kang, Eunuchul; Janjic, Jovana; Noothout, Emile; Chang, Zu-Yao; Verweij, Martin D.; de Jong, Nico; Pertijs, Michiel A. P.

被引用次数: 14

2021

A Readout IC for Capacitive Touch Screen Panels With 33.9 dB Charge-Overflow Reduction Using Amplitude-Modulated Multi-Frequency Excitation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

An, Jae-Sung; Ra, Jong-Hyun; Kang, Eunuchul; Pertijs, Michiel A. P.; Han, Sang-Hyun

被引用次数: 4

2021

Exploiting nonlinear wave propagation to improve the precision of ultrasonic flow meters

ULTRASONICS

Massaad, Jack; van Neer, Paul L. M. J.; van Willigen, Douwe M.; de Jong, Nicolaas; Pertijs, Michiel A. P.; Verweij, Martin D.

被引用次数: 5

2021

Receive/Transmit Aperture Selection for 3D Ultrasound Imaging with a 2D Matrix Transducer

APPLIED SCIENCES-BASEL

Mozaffarzadeh, Moein; Soozande, Mehdi; Fool, Fabian; Pertijs, Michiel A. P.; Vos, Hendrik J.; Verweij, Martin D.; Bosch, Johan G.; de Jong, Nico

被引用次数: 12

2020

Impact of Bit Errors in Digitized RF Data on Ultrasound Image Quality

IEEE TRANSACTIONS ON ULTRASONICS FERROELECTRICS AND FREQUENCY CONTROL

Chen, Zhao; Soozande, Mehdi; Vos, Hendrik J.; Bosch, Johan G.; Verweij, Martin D.; de Jong, Nico; Pertijs, Michiel A. P.

被引用次数: 3

2020

Suppression of Lamb wave excitation via aperture control of a transducer array for ultrasonic clamp-on flow metering

JOURNAL OF THE ACOUSTICAL SOCIETY OF AMERICA

Massaad, Jack; van Neer, Paul L. M. J.; van Willigen, Douwe M.; Pertijs, Michiel A. P.; de Jong, Nicolaas; Verweij, Martin D.

被引用次数: 4

2020

A Variable-Gain Low-Noise Transimpedance Amplifier for Miniature Ultrasound Probes

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kang, Eunuchul; Tan, Mingliang; An, Jae-Sung; Chang, Zu-Yao; Vince, Philippe; Senegond, Nicolas; Mateo, Tony; Meynier, Cyril; Pertijs, Michiel A. P.

被引用次数: 31

2020

A 64-Channel Transmit Beamformer With ± 30 -V Bipolar High-Voltage Pulsers for Catheter-Based Ultrasound Probes

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Tan, Mingliang; Kang, Eunuchul; An, Jae-Sung; Chang, Zu-Yao; Vince, Philippe; Mateo, Tony; Senegond, Nicolas; Pertijs, Michiel A. P.

被引用次数: 27

2020

A 40-nm CMOS Complex Permittivity Sensing Pixel for Material Characterization at Microwave Frequencies

A Pitch-Matched Front-End ASIC With Integrated Subarray Beamforming ADC for Miniature 3-D Ultrasound Probes

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chen, Chao; Chen, Zhao; Bera, Deep; Noothout, Emile; Chang, Zu-Yao; Tan, Mingliang; Vos, Hendrik J.; Bosch, Johan G.; Verweij, Martin D.; de Jong, Nico; Pertijs, Michiel A. P.

被引用次数: 44

2018

A Phase-Domain Readout Circuit for a CMOS-Compatible Hot-Wire CO₂ Sensor

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Cai, Zeyu; van Veldhoven, Robert; Suy, Hilco; de Graaf, Ger; Makinwa, Kofi A. A.; Pertijs, Michiel A. P.

被引用次数: 7

2018

A Front-End ASIC With High-Voltage Transmit Switching and Receive Digitization for 3-D Forward-Looking Intravascular Ultrasound Imaging

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Tan, Mingliang; Chen, Chao; Chen, Zhao; Janjic, Jovana; Daeichin, Varya; Chang, Zu-Yao; Noothout, Emile; van Soest, Gijs; Verweij, Martin D.; de Jong, Nico; Pertijs, Michiel A. P.

被引用次数: 50

2018

A Reconfigurable Ultrasound Transceiver ASIC With 24 x 40 Elements for 3-D Carotid Artery Imaging

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kang, Eunuchul; Ding, Qing; Shabanimotlagh, Maysam; Kruizinga, Pieter; Chang, Zu-Yao; Noothout, Emile; Vos, Hendrik J.; Bosch, Johan G.; Verweij, Martin D.; de Jong, Nico; Pertijs, Michiel A. P.

被引用次数: 30

2018

A 2-D Ultrasound Transducer With Front-End ASIC and Low Cable Count for 3-D Forward-Looking Intravascular Imaging: Performance and Characterization

IEEE TRANSACTIONS ON ULTRASONICS FERROELECTRICS AND FREQUENCY CONTROL

Janjic, Jovana; Tan, Mingliang; Daeichin, Varya; Noothout, Emile; Chen, Chao; Chen, Zhao; Chang, Zu-Yao; Beurskens, Robert H. S. H.; van Soest, Gijs; van der Steen, Antonius F. W.; Verweij, Martin D.; Pertijs, Michiel A. P.; de Jong, Nico

被引用次数: 38

2018

FAST VOLUMETRIC IMAGING USING A MATRIX TRANSESOPHAGEAL ECHOCARDIOGRAPHY PROBE WITH PARTITIONED TRANSMIT-RECEIVE ARRAY

ULTRASOUND IN MEDICINE AND BIOLOGY

Bera, Deep; van den Adel, Franc; Radeljic-Jakic, Nikola; Lippe, Boris; Soozande, Mehdi; Pertijs, Michiel A. P.; Verweij, Martin D.; Kruizinga, Pieter; Daeichin, Varya; Vos, Hendrik J.; van der Steen, Antonius F. W.; Bosch, Johan G.; de Jong, Nico

被引用次数: 5

2018

A Precision Capacitance-to-Digital Converter With 16.7-bit ENOB and 7.5-ppm/°C Thermal Drift

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yang, Ruimin; Pertijs, Michiel A. P.; Nihtianov, Stoyan

被引用次数: 29

2017

A Front-End ASIC With Receive Sub-array Beamforming Integrated With a 32 x 32 PZT Matrix Transducer for 3-D Transesophageal Echocardiography

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chen, Chao; Chen, Zhao; Bera, Deep; Raghunathan, Shreyas B.; Shabanimotlagh, Maysam; Noothout, Emile; Chang, Zu-Yao; Ponte, Jacco; Prins, Christian; Vos, Hendrik J.; Bosch, Johan G.; Verweij, Martin D.; de Jong, Nico; Pertijs, Michiel A. P.

被引用次数: 65

2017

A 7 μ W Offset- and Temperature-Compensated pH-to-Digital Converter

JOURNAL OF SENSORS

Shalmany, Saleh Heidary; Merz, Matthias; Fekri, Ali; Chang, Zu-yao; Hoofman, Romano J. O. M.; Pertijs, Michiel A. P.

被引用次数: 1

2017

A CMOS Readout Circuit for Resistive Transducers Based on Algorithmic Resistance and Power Measurement

IEEE SENSORS JOURNAL

Cai, Zeyu; Guerrero, Luis E. Rueda; Louwerse, Alexander Mattheus Robert; Suy, Hilco; van Veldhoven, Robert; Makinwa, Kofi A. A.; Pertijs, Michiel A. P.

被引用次数: 6

2017

A Ratiometric Readout Circuit for Thermal-Conductivity-Based Resistive CO₂ Sensors

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Cai, Zeyu; van Veldhoven, Robert H. M.; Falepin, Annelies; Suy, Hilco; Sterckx, Eric; Bitterlich, Christian; Makinwa, Kofi A. A.; Pertijs, Michiel A. P.

被引用次数: 20

2016

A Prototype PZT Matrix Transducer With Low-Power Integrated Receive ASIC for 3-D Transesophageal Echocardiography

IEEE TRANSACTIONS ON ULTRASONICS FERROELECTRICS AND FREQUENCY CONTROL

Chen, Chao; Raghunathan, Shreyas B.; Yu, Zili; Shabanimotlagh, Maysam; Chen, Zhao; Chang, Zu-yao; Blaak, Sandra; Prins, Christian; Ponte, Jacco; Noothout, Emile; Vos, Hendrik J.; Bosch, Johan G.; Verweij, Martin D.; de Jong, Nico; Pertijs, Michiel A. P.

被引用次数: 62

2016

A 30 ppm < 80 nJ Ring-Down-Based Readout Circuit for Resonant Sensors

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jiang, Hui; Chang, Zu-yao; Pertijs, Michiel A. P.

被引用次数: 17

2016

Highlights of the ISSCC 2013 Processors and High Performance Digital Sessions

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Fischer, Timothy; Nam, Byeong-Gyu; Chang, Leland; Kuroda, Tadahiro; Pertijs, Michiel A. P.

被引用次数: 0

2014

Low-Power CMOS Smart Temperature Sensor With a Batch-Calibrated Inaccuracy of ± 0.25 °C ($\pm 3\sigma$) from -70 °C to 130 °C

IEEE SENSORS JOURNAL

Aita, Andre L.; Pertijs, Michiel A. P.; Makinwa, Kofi A. A.; Huijsing, Johan H.; Meijer, Gerard C. M.

被引用次数: 65

2013

An Interface for Eddy-Current Displacement Sensors With 15-bit Resolution and 20 MHz Excitation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Nabavi, Mohammad Reza; Pertijs, Michiel A. P.; Nihtianov, Stoyan

被引用次数: 28

2013

A 1.2-V 8.3-nJ CMOS Humidity Sensor for RFID Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Tan, Zhichao; Daamen, Roel; Humbert, Aurelie; Ponomarev, Youri V.; Chae, Youngcheol; Pertijs, Michiel A. P.

被引用次数: 138

2013

Front-End Receiver Electronics for a Matrix Transducer for 3-D Transesophageal Echocardiography

IEEE TRANSACTIONS ON ULTRASONICS FERROELECTRICS AND FREQUENCY CONTROL

Yu, Zili; Blaak, Sandra; Chang, Zu-yao; Yao, Jiajian; Bosch, Johan G.; Prins, Christian; Lancee, Charles T.; de Jong, Nico; Pertijs, Michiel A. P.; Meijer, Gerard C. M.

被引用次数: 39

2012

An Energy-Efficient 15-Bit Capacitive-Sensor Interface Based on Period Modulation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Tan, Zhichao; Shalmany, Saleh Heidary; Meijer, Gerard C. M.; Pertijs, Michiel A. P.

被引用次数: 121

2012

A Single-Temperature Trimming Technique for MOS-Input Operational Amplifiers Achieving 0.33 $\mu\text{V}/^\circ\text{C}$ Offset Drift

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bolatkale, Muhammed; Pertijs, Michiel A. P.; Kindt, Wilko J.; Huijsing, Johan H.; Makinwa, Kofi A. A.

被引用次数: 15

2011

Low-Cost Calibration Techniques for Smart Temperature Sensors

IEEE SENSORS JOURNAL

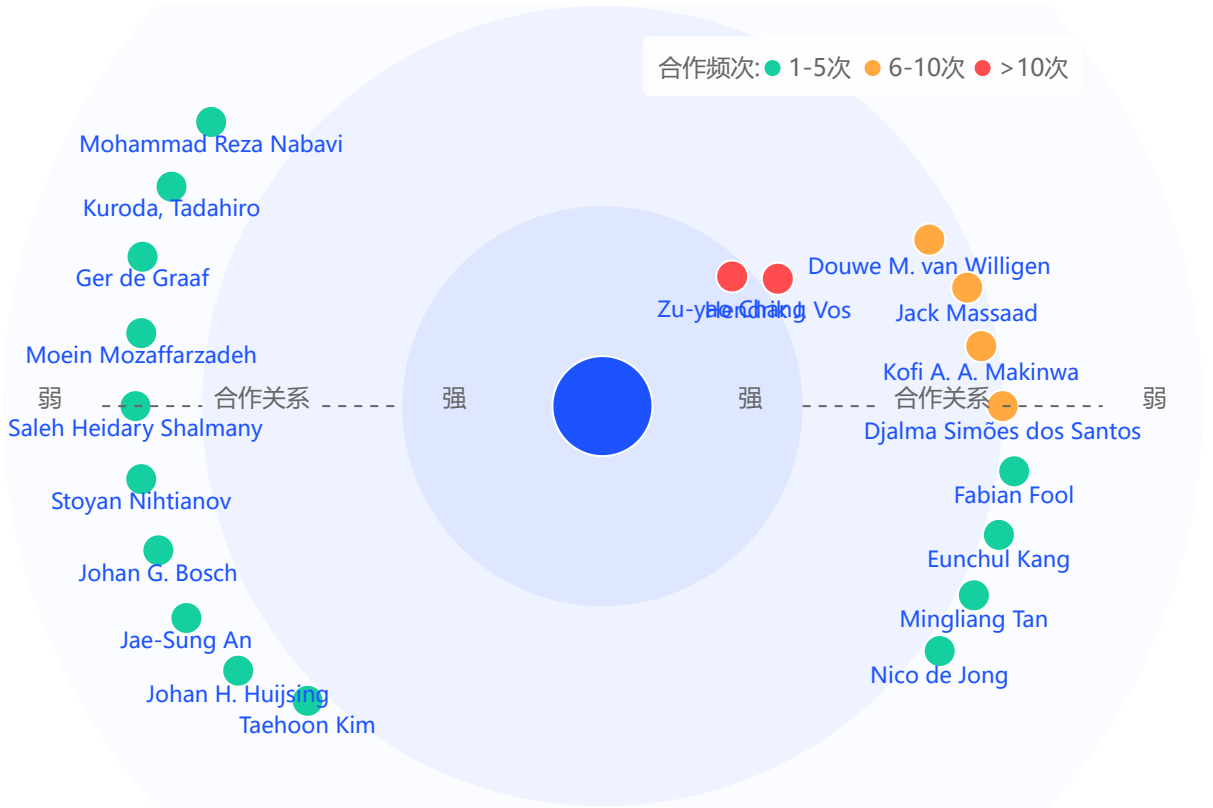
Pertijs, Michiel A. P.; Aita, Andre L.; Makinwa, Kofi A. A.; Huijsing, Johan H.

被引用次数: 41

2010

A Thermal-Diffusivity-Based Frequency Reference in Standard CMOS With an Absolute Inaccuracy of $\pm 0.1\%$ From -55 °C to 125 °C

学术合作



Zu-yao Chang 代尔夫特理工大学	合作次数: 19
Hendrik J. Vos 伊拉兹马斯大学	合作次数: 17
Douwe M. van Willigen 代尔夫特理工大学	合作次数: 8
Jack Massaad 代尔夫特理工大学	合作次数: 7
Kofi A. A. Makinwa 宾夕法尼亚州立大学-帕克	合作次数: 7

还有 17 个合作者